Logic and Proof — Algorithm Worksheet

Run-by-hand problems, hints & solutions for each algorithm in `logic_8020`

Part IB — Cambridge CST — Paper 6

Cover each Solution box and attempt the problem first. Logic & Proof rewards showing every step of these algorithms.

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1 CNF / clause-form conversion

Warm-up 1. Eliminate the connective in $A \to B$.

Hint. $A \to B \equiv \neg A \lor B$.

Solution

$\neg A \lor B$.

Warm-up 2. Put $\neg(A \land B)$ into negation normal form.

Hint. de Morgan.

Solution

$\neg A \lor \neg B$.

Exam-style. Convert $(P \lor Q) \to (R \land S)$ to a clause set; and Skolemise $\forall x\, \exists y\, \forall z\, \exists w\, P(x, y, z, w)$, dropping quantifiers.

Hint. CNF: eliminate $\to$, push $\neg$ in, distribute $\lor$ over $\land$. Skolemise: each $\exists$ under $\forall x_1 \ldots x_k$ becomes a Skolem function $f(x_1, \ldots, x_k)$ of exactly those universally-bound variables.

Solution

$(P \lor Q) \to (R \land S) \equiv \neg(P \lor Q) \lor (R \land S) \equiv (\neg P \land \neg Q) \lor (R \land S)$. Distribute:

$$(\neg P \lor R) \land (\neg P \lor S) \land (\neg Q \lor R) \land (\neg Q \lor S).$$

Clauses: $\{\neg P, R\}, \{\neg P, S\}, \{\neg Q, R\}, \{\neg Q, S\}$.
Skolemisation: $\exists y$ is under $\forall x$ only $\Rightarrow y = f(x)$; $\exists w$ is under $\forall x, \forall z \Rightarrow w = g(x, z)$. Drop the $\forall$s:

$$P\big(x,\ f(x),\ z,\ g(x, z)\big).$$

2 DPLL

Warm-up 1. Apply unit propagation to $\{P, Q\}, \{\neg P\}$.

Hint. The unit $\{\neg P\}$ forces $P = $ false: drop clauses containing $\neg P$, delete P from the rest.

Solution

$\{Q\}$ remains (then it's a unit forcing Q).

Warm-up 2. Find a pure literal in $\{\neg P, Q\}, \{\neg P, R\}, \{\neg Q, \neg R\}$.

Hint. A literal whose complement never appears.

Solution

$\neg P$ is pure (P never occurs positively). Delete the clauses containing it.

Exam-style. Use DPLL to decide $\{A, B\}, \{\neg A, C\}, \{\neg B, C\}, \{\neg C\}$ (i.e. is $(A \vee B) \wedge (A \to C) \wedge (B \to C) \wedge \neg C$ satisfiable?).

Hint. Start with the unit $\{\neg C\}$.

Solution

$\{A, B\}\ \{\neg A, C\}\ \{\neg B, C\}\ \{\neg C\}$	initial
$\{A, B\}\ \{\neg A\}\ \{\neg B\}$	unit $\neg C$
$\{B\}\ \{\neg B\}$	unit $\neg A$ (from $\{A, B\}$ drop A)
$\square$	unit $\neg B \;\Rightarrow\; \{B\}$ empties

Empty clause derived $\Rightarrow$ **unsatisfiable**. Hence $((A \to C) \wedge (B \to C)) \to ((A \vee B) \to C)$ is a theorem.

3 Resolution (propositional)

Warm-up 1. Resolve $\{P, Q\}$ and $\{\neg P, R\}$.

Hint. Cancel the complementary pair $P/\neg P$, union the rest.

Solution

$\{Q, R\}$.

Warm-up 2. Resolve $\{P\}$ and $\{\neg P\}$.

Hint. Nothing left after cancelling.

Solution

The empty clause $\square$ (a contradiction).

Exam-style. Prove by resolution that $\{P \to Q,\ Q \to R,\ P\} \models R$.

Hint. Negate the goal R, convert everything to clauses, resolve to $\square$.

Solution

Clauses: $\{\neg P, Q\}, \{\neg Q, R\}, \{P\}, \{\neg R\}$ (last is the negated goal).

$$\frac{\{P\} \quad \{\neg P, Q\}}{\{Q\}} \qquad \frac{\{Q\} \quad \{\neg Q, R\}}{\{R\}} \qquad \frac{\{R\} \quad \{\neg R\}}{\square}$$

$\square$ reached $\Rightarrow$ the entailment holds.

4 Unification & first-order resolution

Warm-up 1. Unify $f(x, a)$ and $f(b, y)$.

Hint. Unify argument-wise.

Solution

$\{b/x,\ a/y\}$.

Warm-up 2. Factor the clause $\{P(x), P(a)\}$.

Hint. *Unify the two literals.*

Solution

$\sigma = \{a/x\}$ gives the factor $\{P(a)\}$ (a shorter, useful clause).

Exam-style. Use first-order resolution to prove $Q(a)$ from $\forall x\,(P(x) \to Q(x))$ and $P(a)$.

Hint. *Clausify (rename variables apart), negate the goal, resolve with a most general unifier each step.*

Solution

Clauses: $\{\neg P(x), Q(x)\}$, $\{P(a)\}$, $\{\neg Q(a)\}$ (negated goal).

$$\frac{\{P(a)\}\quad\{\neg P(x), Q(x)\}}{\{Q(a)\}}\ \sigma = \{a/x\} \qquad \frac{\{Q(a)\}\quad\{\neg Q(a)\}}{\square}$$

$\square \Rightarrow Q(a)$ follows. (Always rename apart before resolving, and use the *occurs check* in unification.)

5 Binary Decision Diagrams

Warm-up 1. Give the ROBDD of the single variable P, and of $\neg P$.

Hint. *One decision node; negation just swaps the leaves.*

Solution

P: node on P with true-child $\mathbf{1}$, false-child $\mathbf{0}$. $\neg P$: same node with leaves swapped ($\mathbf{0}/\mathbf{1}$).

Warm-up 2. When is a decision node deleted during reduction?

Hint. *Irredundancy.*

Solution

When its true- and false-children are identical (the test is irrelevant) — replace the node by that common child.

Exam-style. Build the ROBDD for $(P \wedge Q) \vee (\neg P \wedge R)$ with ordering $P < Q < R$. Then state how you'd use BDDs to check whether two formulas are equivalent.

Hint. *This is "if P then Q else R" — a multiplexer. Test P first; on each branch you're left with a single remaining variable.*

Solution

$$\underbrace{P}_{\text{root}}: \quad \text{if } P \to (\,Q : \mathbf{1}/\mathbf{0}\,), \quad \text{else} \to (\,R : \mathbf{1}/\mathbf{0}\,).$$

i.e. node P; its true-branch is a node on Q (true$\to$ $\mathbf{1}$, false$\to$ $\mathbf{0}$); its false-branch is a node on R (true$\to$ $\mathbf{1}$, false$\to$ $\mathbf{0}$). Already reduced: the Q- and R-nodes have distinct children, and P's two children differ.

Equivalence check: build both ROBDDs under the *same* variable ordering; they are logically equivalent *iff* the two diagrams are identical (with hashing/sharing, iff their root pointers are equal). A tautology reduces to $\mathbf{1}$, an unsatisfiable formula to $\mathbf{0}$.

6 Fourier–Motzkin variable elimination

Warm-up 1. Eliminate x from $\{x \leq 5,\ x \geq 2\}$.

Hint. *Pair the lower bound with the upper bound.*

Solution

$2 \le 5$ (true) $\Rightarrow$ satisfiable ($x \in [2, 5]$).

Warm-up 2. Eliminate x from $\{x \le 3,\ x \ge 7\}$.

Hint. *Same, but check the resulting inequality.*

Solution

$7 \le 3$ (false) $\Rightarrow$ unsatisfiable.

Exam-style. Decide satisfiability of $\{\, x \le y,\ y \le z,\ z \le x - 1\,\}$.

Hint. *Rewrite as lower/upper bounds and eliminate one variable at a time; a cyclic chain should produce a contradiction.*

Solution

From $z \le x - 1$: lower bound $x \ge z + 1$; upper bound $x \le y$. Eliminate x: $z + 1 \le y$. Now $\{z + 1 \le y,\ y \le z\}$. Eliminate y: lower $y \ge z + 1$, upper $y \le z \Rightarrow z + 1 \le z \Rightarrow 1 \le 0$. **Contradiction** $\Rightarrow$ **unsatisfiable** (the constraints say $x \le y \le z \le x - 1$, impossible).

7 Sequent calculus

Warm-up 1. Is the sequent $A \Rightarrow A$ provable?

Hint. *Basic sequent.*

Solution

Yes — it is a *basic* (axiom) sequent: the same formula on both sides.

Warm-up 2. Which rule reduces $\Gamma \Rightarrow \Delta,\ A \wedge B$, and how many premises does it have?

Hint. *Right rule for* $\wedge$.

Solution

$(\wedge r)$, with *two* premises: $\Gamma \Rightarrow \Delta, A$ and $\Gamma \Rightarrow \Delta, B$.

Exam-style. Give backward sequent-calculus proofs of (a) $\Rightarrow (A \wedge B) \to (B \wedge A)$ and (b) $\Rightarrow \forall x\,(P(x) \wedge Q(x)) \to \forall x\, P(x)$.

Hint. *(a)* $(\to r)$ *then* $(\wedge l)$ *and* $(\wedge r)$. *(b) Apply* $(\to r)$, *then* $(\forall r)$ *before* $(\forall l)$ *(the* $\forall r$ *proviso "x not free in the conclusion" must hold), then* $(\wedge l)$.

Solution

(a)

$$\cfrac{\cfrac{\cfrac{A, B \Rightarrow B, A}{A \wedge B \Rightarrow B, A}\,(\wedge l) \quad \text{and} \quad \cfrac{A, B \Rightarrow A}{\dots}}{A \wedge B \Rightarrow B \wedge A}\,(\wedge r)}{\Rightarrow (A \wedge B) \to (B \wedge A)}\,(\to r)$$

i.e. $(\to r)$ moves $A \wedge B$ left; $(\wedge l)$ splits it into A, B; $(\wedge r)$ splits the goal $B \wedge A$ into two basic sequents $A, B \Rightarrow B$ and $A, B \Rightarrow A$.

(b)

$$\dfrac{\dfrac{\dfrac{\dfrac{P(x), Q(x) \Rightarrow P(x)}{P(x) \wedge Q(x) \Rightarrow P(x)}\ (\wedge l)}{\forall x(P(x) \wedge Q(x)) \Rightarrow P(x)}\ (\forall l)}{\forall x(P(x) \wedge Q(x)) \Rightarrow \forall x\, P(x)}\ (\forall r)}{\Rightarrow \forall x(P(x) \wedge Q(x)) \to \forall x\, P(x)}\ (\to r)$$

Crucially $(\forall r)$ is applied first (introducing a fresh x not free below), then $(\forall l)$ instantiates the assumption at that same x; $(\wedge l)$ then gives a basic sequent.

8 Modal S4 sequent proof

Warm-up 1. State the $(\Box r)$ rule, including its side condition.

Hint. It discards formulas that don't begin with the right modality.

Solution

$\dfrac{\Gamma^* \Rightarrow \Delta^*, A}{\Gamma \Rightarrow \Delta, \Box A}$ $(\Box r)$, where $\Gamma^* = \{\Box B \in \Gamma\}$ and $\Delta^* = \{\Diamond B \in \Delta\}$ — all non-$\Box$ left formulas and non-$\Diamond$ right formulas are thrown away.

Exam-style. Prove the distribution axiom $\Box(A \to B),\ \Box A \ \Rightarrow\ \Box B$ in S4.

Hint. Apply $(\Box r)$ first (so the $\Box$ assumptions survive into Γ^*); then strip the boxes with $(\Box l)$ and finish with $(\to l)$.

Solution

$$\dfrac{\dfrac{\dfrac{\overline{A \Rightarrow A, B} \quad \overline{B, A \Rightarrow B}}{A \to B,\ A \Rightarrow B}\ (\to l)}{\Box(A \to B), \Box A \Rightarrow B}\ (\Box l\ (\times 2))}{\Box(A \to B), \Box A \Rightarrow \Box B}\ (\Box r)$$

Reading top-down: $(\Box r)$ reduces the goal $\Box B$ to B, keeping $\Gamma^* = \{\Box(A \to B), \Box A\}$ (both begin with $\Box$, so they survive); $(\Box l)$ strips each box to give $A \to B,\ A \Rightarrow B$; $(\to l)$ splits into the two basic sequents $A \Rightarrow A, B$ and $B, A \Rightarrow B$.

Re-run DPLL, resolution, a BDD, Fourier–Motzkin, and one sequent proof cold before the exam.